

TRAVELING WAVES FOR THE PERTURBED NON-LINEAR
SCHRÖDINGER'S EQUATION

A Thesis
Submitted to the Graduate School
of
Tennessee State University
in
Partial Fulfillment of the Requirements
for the Degree of
Master of Science

Graduate Research Series No. 2278

Davon Waheed Williams

August 2005

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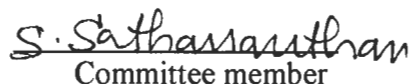
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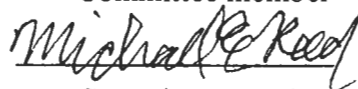
We are submitting a thesis written by Davon Waheed Williams entitled "Traveling Waves for the Perturbed Non-Linear Schrodinger's Equation". We recommend that it be accepted in partial fulfillment of the requirements for the degree, Masters of Science in Mathematical Science.



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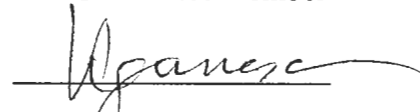
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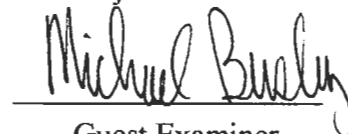
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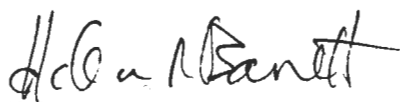


Major Advisor



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Accepted for the Graduate School



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DEDICATION

I dedicate this thesis report project to my mother Debra Cox Hamilton, my loving grandfather and grandmother Carlos and Corene Cox my Aunt and Uncle Brian and Claudine Smith and all my loving family members who have faithfully supported and encouraged me throughout my academic career.

D.W.W

ABSTRACT

DAVON WAHEED WILLIAMS. Traveling Waves for the Perturbed Non-Linear Schrödinger's Equation (under the direction of **DR. ANJAN BISWAS**).

The objective of this research was to investigate the effects of small perturbations on the propagation of the traveling waves for the perturbed non-linear Schrödinger's equation. In our work we were interested in finding a traveling wave solution for this equation. In everyday places such as local airports, ship docks and police facilities wave patterns and signals are used. Also in the world of computers and telecommunications the use of waves plays a vital role. In order to use the internet or simply to make a phone call wave or wave signals make it all possible.

ACKNOWLEDGEMENT

First and foremost, I give thanks to Almighty God for giving me the guidance, ability and strength to successfully accomplish this thesis report project.

This project was written under the direction and supervision of Dr. Anjan Biswas. I thank him for his knowledge that was shared with me unselfishly; his valuable time, which he gave me so generously; patience that he demonstrated day after day; and also his enthusiasm, which was contagious.

Special thanks to Dr. Kothandaraman Ganesan, Graduate Program Coordinator in the Department of Physics and Mathematics at Tennessee State University, who provided the academic opportunity and advisement. Also thanks to the Center of Excellence Information Systems and the Mathematics Department for financial support. I would also like to thank Dr. Sandra H. Scheick, the Head of Physics and Mathematics Department at Tennessee State University, for her encouragement and support. A special thanks to Mrs. Debra Alexander Secretary, Physics and Mathematics Department at Tennessee State University, for her patience and support.

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CHAPTER I

INTRODUCTION

Self-action and interaction of wave packets in nonlinear media with periodically varying dispersion or diffraction is a fundamental physical problem with a wide range of practical applications [1]. Optical pulse transmission in dispersion managed optical fibers, stretched pulse generation in mode-locked laser systems and recirculating fiber loops, propagation of guiding-center solitons and light bullets in nonlinear media, evolution of soliton like beams in periodically modulated cubic nonlinear media and in Bose-Einstein condensates [3].

During the last decade, impressive progress was achieved in this area from theoretical as well as experimental and practical points of view. Several methods have been developed to study the propagation of traveling solitary nonlinear waves in dispersion or diffraction managed systems. The different methods are variational approaches, the guiding-center concept, the multiscale theory, spectral domain analysis, and the numerical averaging method. In addition to conventional bright dispersion managed solitons, dark, gray, and antisymmetric solitons were also studied recently [6]. Dispersion management offers a variety of advantages of paramount practical importance in high-bitrate, long-haul optical fiber links [4, 5]. In this context, the key features of dispersion-managed systems employing so-called return-to-zero, or soliton like pulse trains, are the reduced pulse-pulse interaction and the pulse energy enhancement, which

are possible under proper conditions. This reduces the inter-symbol-interaction errors and enhances the signal-to-noise ratio; hence dispersion-managed soliton pulse trains exhibit improved performances in data transmission communications [2, 3]. Thus, high-bit-rate, long-haul, all-optical fiber links employing this technique are already installed and in use. Besides localized single-pulse solutions, the nonlinear Schrödinger equation (NLSE) serves as a starting point for the description of many dispersion-managed systems. Periodic solutions in the form of Jacobi elliptic functions, lead to so-called cn , dn , and sn waves which are stable multicolor periodic-wave arrays and exist in different regimes characterized by a varying degree of localization. In the regime of strong localization, dn and cn waves feature, respectively, trains of in-phase or out-of phase well-separated soliton like pulses. Similarly, sn waves describe trains of out-of-phase dark solitons or optical links. In the case of weak localization, the traveling wave concept bridges the gap between soliton and linear harmonic cn and sn cases or continuous dn case waves. From this point of view, periodic nonlinear waves play a similar role in nonlinear optics as diffraction gratings in classical linear optics [4].

On the other hand, passive mode-locked fiber lasers under certain conditions can deliver a continuous wave train of picosecond or femtosecond pulses at a repetition rate higher than 100 GHz, with essentially discrete Fourier spectra of synchronized modes, which can be effectively described by traveling waves [4, 5, 7]. The properties of stationary traveling waves supported by the NLSE in uniform media have been studied in detail. Nevertheless, the impact of dispersion management on the traveling wave pulse

trains, hence the corresponding new features and practical advantages, have never been addressed to date.

1.1 The Purpose of the Research

The objective of this research is to investigate the effects of small perturbations on the propagation of traveling waves by using mathematical analysis to determine the approximate behavior of perturbed traveling wave.

1.2 Organization of this Project Report

Chapter one of this thesis introduces the background of this project. Chapter two discusses the fundamental knowledge about traveling waves and gives a review on elliptic functions. Chapter three presents a derivation of the Nonlinear Schrödinger's equation as related to traveling waves. Finally the method of solving a given traveling wave for the perturbed nonlinear Schrödinger's equation and conclusions are elaborated in chapter four.

CHAPTER II

TRAVELING WAVES AND ELLIPTIC FUNCTIONS

2.1. Introduction of Traveling Waves

Traveling waves are non-linear waves that are obtained by wave ansatz for nonlinear evolution equations. A traveling wave is given by

$$u(x,t) = f(x - ct) \quad (2.1)$$

where x and t are independent variables and c is the velocity of the wave. This functional form hypothesis $f(x - ct)$ leads to solutions in terms of elliptic functions [2].

Solutions of the dissipative system, starting near a traveling wave, are approximated with a long curve on the family of traveling waves. The approximation process is sharp in the sense that as a solution tends to zero as t approaches infinity, the difference between the solution and the approximation tends to zero in a norm that sharply picks out their difference in shape [2].

2.2 Elliptic Functions- A Review

Consider the integral

$$w = \int_0^\phi \frac{d\theta}{(1 - m \sin^2 \theta)^{\frac{1}{2}}}, \quad (2.2)$$

It can be observed that integral (2.2) defines the inverse of the trigonometric function $\sin$. This led Jacobi and Abel to define a new pair of inverse functions (2.4) from (2.3)

$$v = \int_0^\phi \frac{d\theta}{(1 - m \sin^2 \theta)^{\frac{1}{2}}} \quad (2.3)$$

$$\operatorname{sn}\phi = \sin \phi, \quad \operatorname{cn}\phi = \cos \phi \quad (2.4)$$

There are two of the Jacobian elliptic functions. They are usually written as

$$\operatorname{sn}(\phi|m), \quad \text{and} \quad \operatorname{cn}(\phi|m)$$

to denote the dependence on the parameter. (it is not unusual to work with the modulus, k , where $m = k^2$; however, m is slightly more advantageous in the present context) [2].

Jacobi and Abel considered the case when $0 < m < 1$

The two special cases $m = 0, 1$ enable integrals (2.3) and functions (2.4) to be reduced to elementary functions

Case I : If $m = 0$

$$\text{then, } \phi = \varphi \quad \text{and so} \quad \operatorname{cn}(\varphi|0) = \cos \varphi,$$

Case II: If $m = 1$

then the integral (2.3) can be evaluated to yield

$$\phi = \operatorname{sech}^{-1}(\cos \phi) \quad \text{and so} \quad \operatorname{cn}(\phi|1) = \operatorname{sech} \phi$$

It therefore follows that $\operatorname{cn}(\phi|m)$ and $\operatorname{sn}(\phi|m)$ are periodic functions for

$0 \leq m \leq 1$, but that periodicity is lost for $m = 1$. Now the period of cn and sn corresponds to the period 2π of cosine and sine, and so the period of these elliptic functions can be

written as

$$\int_0^{2\pi} \frac{d\theta}{(1 - m \sin^2 \theta)^{\frac{1}{2}}} = 4 \int_0^{\pi/2} \frac{d\theta}{(1 - m \sin^2 \theta)^{\frac{1}{2}}}$$

This latter integral is the complete elliptic integral of the first kind ('complete' because it has fixed limits),

$$K(m) = \int_0^{\pi/2} \frac{d\theta}{(1 - m \sin^2 \theta)^{\frac{1}{2}}} \quad (2.5)$$

It is immediately clear that $K(0) = \pi/2$, and it is also straightforward to show that $K(m)$ increases monotonically as m increases. In fact

$$K(m) \sim \frac{1}{2} \log\{16/(1-m)\} \text{ as } m \rightarrow 1^-,$$

and so $K(m) \rightarrow +\infty$ as $m \rightarrow 1^-$. Of course, this just demonstrates the infinite 'period' of the $\text{cn}(\phi|m) = \text{sech } \phi$

Algebraic and differential relations between the Jacobian elliptic functions can also be obtained quite easily. For example

$$\text{cn}^2 + \text{sn}^2 = 1$$

(from the Pythagorean result for the trigonometric functions), and

$$\frac{d}{d\phi} \text{cn} = \frac{d\phi}{d\phi} \frac{d}{d\phi} \text{cn} = (1 - m \sin^2 \phi)^{1/2} \frac{d}{d\phi} \cos \phi$$

which is usually expressed in terms of a elliptic function

$$\text{dn}(\phi|m) = (1 - m \sin^2 \phi)^{1/2},$$

$$\text{so that } \frac{d}{d\phi} \text{cn} = -\text{sn } \text{dn}$$

(and also it can be observed that $\text{dn}^2 + m \text{sn}^2 = 1$). It can be noted that, since the argument (ϕ) and parameter (m) are the same throughout these identities, they have been suppressed altogether.

CHAPTER III

THE NONLINEAR SCHRÖDINGER EQUATION

We derive the nonlinear Schrödinger equation that describes the behavior of solitons in optical fibers, starting from first principles, i.e. Maxwell's equations in isotropic, dispersive, and nonlinear dielectric material. To avoid complexity in computations, we restrict the derivation to the propagation of wave packets in a planar wave guide, where the electric field is taken to be a scalar property. Despite restrictions, the essential idea is the same.

3.1 Maxwell Equations

We begin with Maxwell's equations

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \quad (3.1)$$

$$\nabla \times \vec{B} = \mu_0 \frac{\partial \vec{D}}{\partial t}, \quad (3.2)$$

$$\nabla \cdot \vec{D} = 0, \quad (3.3)$$

$$\nabla \cdot \vec{B} = 0, \quad (3.4)$$

where E , D , B represents electric field, electric flux density, and magnetic flux density, respectively. Since the magnetic properties of silica can be ignored, the permeability is given by μ_0 , namely the one for free space. Eliminating B in (3.1)-(3.4), we obtain

$$\nabla \times (\nabla \times \vec{E}) = -\frac{\partial}{\partial t}(\nabla \times \vec{B}) = -\mu_0 \frac{\partial^2 \vec{D}}{\partial t^2} \quad (3.5)$$

In dielectric material, $\vec{D}$ is expected as the sum of the contributions of the electric field in free space $\epsilon_0 \vec{E}$ and the effect of the electric polarization of the material $\vec{P}$:

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}, \quad (3.6)$$

where ϵ_0 is the permittivity of free space. Silica exhibits linear and nonlinear response with retardation. The electric polarization $\vec{P}$ has thus two contributions:

$$\begin{aligned} \vec{P} &= \vec{P}_L + \vec{P}_{NL} \\ &= \epsilon_0 \int_{-\infty}^t \chi^{(1)}(r, t - \tau) \cdot E(r, \tau) d\tau \\ &\quad + \epsilon_0 \int_{-\infty}^t \int_{-\infty}^t \int_{-\infty}^t \chi^{(3)}(r, t - \tau_1, t - \tau_2, t - \tau_3) : E(r, \tau_1) E(r, \tau_2) E(r, \tau_3) d\tau_1 d\tau_2 d\tau_3, \end{aligned} \quad (3.7)$$

where $\chi^{(1)}$ and $\chi^{(3)}$ represent the linear and third order nonlinear susceptibility tensors, respectively, having the dependence of the spatial coordinates r of the material, and

$$\begin{aligned} (\chi^{(1)} \cdot E)_i &= \sum_j \chi_{ij}^{(1)} E_j \\ (\chi^{(3)} : EEE)_i &= \sum_{j,k,l} \chi_{ijkl}^{(3)} E_j E_k E_l \\ (i = x, y, z). \end{aligned} \quad (3.8)$$

Because of the centrosymmetric crystal structure of silica, optical fibers do not exhibit the second order nonlinear effect. Using the following formula of the vector calculus

$$\nabla \times \nabla \times \vec{E} = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} \quad (3.9)$$

on the left hand side of (3.5), and assuming a weakly guided mode (i.e. $\nabla \cdot \vec{E} = 0$), we have

$$\nabla^2 \vec{E} - \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2} = \frac{1}{c^2 \epsilon_0} \frac{\partial^2 \vec{P}}{\partial t^2} \quad (3.10)$$

$$\begin{aligned} &= \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \int_{-\infty}^t \chi^{(1)}(r, t-\tau) \cdot E(r, \tau) d\tau \\ &+ \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \int_{-\infty}^t \int_{-\infty}^t \int_{-\infty}^t \epsilon_0 \chi^{(3)}(r, t-\tau_1, t-\tau_2, t-\tau_3) : E(r, \tau_1) E(r, \tau_2) E(r, \tau_3) d\tau_1 d\tau_2 d\tau_3, \end{aligned} \quad (3.11)$$

3.2 Quasi monochromatic approximation

We introduced two assumptions to derive the nonlinear Schrödinger equation by means of a proper approximation of (3.10). First, we assume the electric field to be a scalar quantity, since the planer wave guide allows a single transverse polarization, unlike the case of the guided modes in optical fibers. Second, we apply the quasi monochromatic approximation, namely we assume that the waveform having slowly varying envelope, or in the frequency domain having spectrum with respect to its carrier frequency. The latter approximation leads to slow variation of the light wave envelope with respect to time and distance. The nonlinear Schrödinger equation is derived as the equation that describes long range evolution of the slowly varying envelope. Before introducing the quasi-monochromatic approximation, we first write down the electric field vector E as a plane wave in the form

$$\begin{aligned} E(x, z, t) &= \hat{y} E(x, z, t) \\ &= \hat{y} [A U(x, \omega_0) \exp(ik_0 z - i\omega_0 t) + c.c.], \end{aligned} \quad (3.12)$$

where c.c. denotes the complex conjugate, $\hat{y}$ is the unit vector along y axis, ω_0 and

$k_0 \equiv k(\omega_0)$ represents the angular frequency and the wave number of the carrier respectively, and A and U are the wave envelope and the mode distribution over a cross section in the wave guide geometry. We assumed the waveguide having guiding structure in x and y coordinates, namely optical wave propagates along z axis, and having uniform structure in y direction (i.e. $\partial/\partial y = 0$) so that E is polarized along the y axis. Taking the Fourier transform of (3.11), we obtain the electric field in frequency domain:

$$\begin{aligned}\hat{E}(x, z, \omega) &= \int_{-\infty}^{\infty} E(x, z, t) \exp(i\omega t) dt \\ &= 2\pi\delta(\omega - \omega_0)AU(x, \omega_0) \exp(ik_0 z) + c.c.\end{aligned}\tag{3.13}$$

Note that A is constant with respect to t and z at present, as long as the wave is monochromatic, i.e. the spectrum is given by the delta function standing at the frequency $\omega = \pm\omega_0$. However, as a result of the quasi-monochromatic approximation, the envelope A acquires sufficiently slow variation in time domain, whereas in frequency domain it exhibits sufficiently small spectral broadening over the bandwidth $\Delta\omega$ in the vicinity of $\pm\omega_0$. We introduced a small parameter $\varepsilon \sim \Delta\omega / \omega_0$ as a factor characterizing the small scale of the spectral broadening. Long range evolution of A is described in a slow scale coordinate $Z = \varepsilon z$. As a consequence of the long term amplitude modulation, $\hat{E}(x, z, \omega)$ now has the form

$$\hat{E}(x, z, \omega) = \hat{A}(Z, \omega - \omega_0)U(x, \omega_0) \exp(ik_0 z) + c.c.\tag{3.14}$$

By taking the inverse Fourier transform, we have

$$\begin{aligned}
& \hat{E}(x, z, \omega) \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{E}(x, z, t) \exp(-i\omega t) dt \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{A}(Z, \omega - \omega_0) U(x, \omega_0) \exp(ik_0 z - i\omega t) + c.c. \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{A}(Z, \omega - \omega_0) \exp[-i(\omega - \omega_0)t] d(\omega - \omega_0) U(x, \omega_0) \exp(ik_0 z - i\omega t) + c.c. \\
&= \frac{\varepsilon}{2\pi} \int_{-\infty}^{\infty} \hat{A}(Z, \varepsilon\Omega) \exp(-i\Omega T) d\Omega U(x, \omega_0) \exp(ik_0 z - i\omega_0 t) + c.c. \\
&= \varepsilon A(Z, T; \varepsilon) U(x, \omega_0) \exp(ik_0 z - i\omega_0 t) + c.c.
\end{aligned} \tag{3.15}$$

where we introduced another independent variable $T = \varepsilon t$ similar to Z to describe long term behavior of A , and

$$A(Z, T; \varepsilon) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{A}(Z, \Omega; \varepsilon) \exp(-i\Omega T) d\Omega, \varepsilon\Omega \equiv \omega - \omega_0. \tag{3.16}$$

3.3 Linear and Nonlinear polarization

We impose the quasi-monochromatic approximation to the linear and nonlinear polarization in the scalar Maxwell equation

$$\begin{aligned}
& \frac{\partial^2 \bar{E}}{\partial x^2} + \frac{\partial^2 \bar{E}}{\partial z^2} - \frac{1}{c^2} \frac{\partial^2 \bar{E}}{\partial t^2} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \int_{-\infty}^t \chi^{(1)}(x, t - \tau) \bar{E}(x, z, \tau) d\tau \\
& - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \iiint \chi^{(3)}(x, t - \tau_1, t - \tau_2, t - \tau_3) E(x, z, \tau_1) E(x, z, \tau_2) E(x, z, \tau_3) d\tau_1 d\tau_2 d\tau_3 = 0
\end{aligned} \tag{3.17}$$

where $x^{(1)} = x_{yy}^{(1)}$ and $x^{(3)} = x_{yyy}^{(3)}$ are scalar functions of x and t and assumed to be

independent of z . Since we are interested in the behavior of lightwave electric fields in the vicinity of the stationary state ($\varepsilon = 0$), it is natural to introduce an asymptotic expansion of $E(x, z, t)$ with respect to ε :

$$E(x, z, t) = E_0 + \varepsilon E_1 + \varepsilon^2 E_2 + \dots \tag{3.18}$$

Note that, in the frame of the quasi-monochromatic approximation, $E_i \sim O(\varepsilon)$ as shown in (3.14). We apply the quasi-monochromatic approximation to the linear and the nonlinear polarization for each $O(\varepsilon^n)$. First we consider $O(\varepsilon)$ contributions in the linear polarization term. Taking the Fourier transform of the $O(\varepsilon)$ component in P_L , we obtain

$$\begin{aligned} & F.T. \left(\frac{\partial^2}{\partial t^2} \int \chi^{(1)}(x, t - \tau) E_0(x, z, \tau) d\tau \right) \\ &= -\omega^2 \chi^{(1)}(x, \omega) \hat{E}_0(x, z, \omega) \\ &= -\omega^2 \hat{\chi}^{(1)}(x, \omega) \hat{A}(Z, \omega - \omega_0) U(x, \omega_0) \exp(ik_0 z) + c.c., \end{aligned} \quad (3.19)$$

Following (3.14), the inverse transform (i.e. multiply $(1/2\pi)\exp(-i\omega t)$ and integrate over ω) results in

$$\begin{aligned} & \frac{\partial^2 \Omega}{\partial u^2} \int \chi^{(1)}(x, t - \tau) E_0(x, z, \tau) d\tau \\ &= -\frac{1}{2\pi} \int \omega^2 \hat{\chi}^{(1)}(x, \omega) \hat{A}(Z, \omega - \omega_0) U(x, \omega_0) \exp(ik_0 z - i\omega t) d\omega + c.c. \\ &= -\left(\frac{1}{2\pi} \int [\omega_0 + (\omega - \omega_0)]^2 \hat{\chi}^{(1)}(x, \omega_0 + (\omega - \omega_0)) \hat{A}(Z, \omega - \omega_0) \exp[-i(\omega - \omega_0)t] d(\omega - \omega_0) \right) \\ & \quad \times U(x, \omega_0) \exp(ik_0 z - i\omega_0 t) + c.c. \\ &= -\left(\frac{\varepsilon}{2\pi} \int (\omega_0 + \varepsilon\Omega)^2 \hat{\chi}^{(1)}(x, \omega_0 + \varepsilon\Omega) \hat{A}(Z, \varepsilon\Omega) \exp(-i\Omega T) d\Omega \right) U(x, \omega_0) \exp(ik_0 z - i\omega t) + c.c. \\ &= -\varepsilon \left(\omega_0 + i\varepsilon \frac{\partial}{\partial T} \right)^2 \chi^{(1)}(x, \omega_0 + i\varepsilon \frac{\partial}{\partial T}) \hat{A}(Z, T; \varepsilon) U(x, \omega_0) \exp(ik_0 z - i\omega t) + c.c., \end{aligned} \quad (3.20)$$

where we referred to the following relation

$$(i\varepsilon \frac{\partial}{\partial T})^n A(Z, T; \varepsilon) = \frac{\varepsilon}{2\pi} \int \varepsilon^n \Omega^n \hat{A}(Z, \Omega, \varepsilon) \exp(-i\Omega T) d\Omega. \quad (3.21)$$

It can be easily verified that the higher order terms of P_L have a form similar to (3.19).

Kerr nonlinearity has a lowest order contribution at $O(\varepsilon^3)$. The quasi-monochromatic

approximation to the nonlinear electric polarization P_{NL} requires somewhat complicated calculations, but in principal, we just follow the calculations for the linear polarization.

We first take the Fourier transform of the nonlinear term:

$$\begin{aligned}
& F.T. \left(\frac{\partial^2}{\partial t^2} \iiint \hat{\chi}^{(3)}(x, t - \tau_1, t - \tau_2, t - \tau_3) E_0(x, z, \tau_1) E_0(x, z, \tau_2) E_0(x, z, \tau_3) d\tau_1 d\tau_2 d\tau_3 \right) \\
&= F.T. \left(-\frac{1}{(2\pi)^3} \iiint d\omega_1 d\omega_2 d\omega_3 (\omega_1 + \omega_2 + \omega_3)^2 \hat{\chi}^{(3)}(x, \omega_1, \omega_2, \omega_3) \right. \\
&\quad \times \hat{E}_0(x, z, \omega_1) \exp(-i\omega_1 t) \hat{E}_0(x, z, \omega_2) \exp(-i\omega_2 t) \hat{E}_0(x, z, \omega_3) \exp(-i\omega_3 t) \Big) \\
&= -\frac{1}{(2\pi)^3} \int dt \exp(i\omega t) \iiint d\omega_1 d\omega_2 d\omega_3 (\omega_1 + \omega_2 + \omega_3)^2 \chi^{(3)}(x, \omega_1, \omega_2, \omega_3) \\
&\quad \times \hat{E}_0(x, z, \omega_1) \hat{E}_0(x, z, \omega_2) \hat{E}_0(x, z, \omega_3) \exp(-i(\omega_1 + \omega_2 + \omega_3)t) \Big) \\
&= -\frac{1}{(2\pi)^3} \iiint d\omega_1 d\omega_2 d\omega_3 2\pi \delta(\omega - \omega_1 - \omega_2 - \omega_3) (\omega_1 + \omega_2 + \omega_3)^2 \hat{\chi}^{(3)}(x, \omega_1, \omega_2, \omega_3) \\
&\quad \times \hat{E}_0(x, z, \omega_1) \hat{E}_0(x, z, \omega_2) \hat{E}_0(x, z, \omega_3) \\
&= -\frac{1}{(2\pi)^2} \iiint d\omega_1 d\omega_2 d\omega_3 \delta(\omega - \omega_1 - \omega_2 - \omega_3) (\omega_1 + \omega_2 + \omega_3)^2 \hat{\chi}^{(3)}(x, \omega_1, \omega_2, \omega_3) \\
&\quad \times (\hat{A}(Z, \omega_1 - \omega_0) U(x, \omega_0) \exp(ik_0 z) + c.c.) (\hat{A}(Z, \omega_2 - \omega_0) U(x, \omega_0) \exp(ik_0 z) + c.c.) \\
&\quad \times (\hat{A}(Z, \omega_3 - \omega_0) U(x, \omega_0) \exp(ik_0 z) + c.c.), \tag{3.22}
\end{aligned}$$

where we used the formula

$$\int_{-\infty}^{\infty} \exp[i(\omega - \omega_1 - \omega_2 - \omega_3)t] dt = 2\pi \delta(\omega - \omega_1 - \omega_2 - \omega_3) \tag{3.23}$$

in the integration over t in (3.21). We note that the nonlinear term (3.21) includes eight contributions in total: three components having the wave number of k_0 and $-k_0$ respectively i.e. $\alpha \exp(ik_0 z)$ and $(\alpha \exp(-ik_0 z))$, and one harmonic component having the wave number of $3k_0$ and $-3k_0$ respectively one of the $\alpha \exp(3ik_0 z)$ and

$\alpha \exp(-3ik_0 z)$. We are interested in $\exp(\pm ik_0 z)$ terms; we consider specifically one of the $\exp(ik_0 z)$ components of the form

$$= -\frac{1}{(2\pi)^2} \iiint d\omega_1 d\omega_2 d\omega_3 \delta(\omega - \omega_1 - \omega_2 - \omega_3) (\omega_1 + \omega_2 + \omega_3)^2 \hat{\chi}^{(3)}(x, \omega_1, \omega_2, \omega_3) \quad (3.24)$$

$$\times (\hat{A}^*(Z, -\omega_1 - \omega_0) \hat{A}(Z, \omega_2 - \omega_0) \hat{A}(Z, \omega_3 - \omega_0) U^*(x, \omega_0) U(x, \omega_0) U(x, \omega_0) \exp(ik_0 z))$$

among the three contributions. Taylor expansion in the vicinity of $\omega_1 = -\omega_0$, ($\omega_2 = \omega_0$)

and $\omega_3 = \omega_0$ in (3.23) yields

$$-\frac{1}{(2\pi)^2} \iiint d\omega_1 d\omega_2 d\omega_3 \delta(\omega - \omega_1 - \omega_2 - \omega_3)$$

$$\times (\omega_0^2 + 2\omega_0(\omega_1 - (\omega_0)^2 + 2\omega_0(\omega_2 - (\omega_0)^2 + 2\omega_0(\omega_3 - (\omega_0)^2 + \dots)$$

$$\times (\hat{\chi}^{(3)}(x, -\omega_0, \omega_0, \omega_0)(\omega_2 - \omega_0) + \frac{\partial \hat{\chi}^{(3)}}{\partial \omega_1}(x, -\omega_0, \omega_0, \omega_0)(\omega_1 - (-\omega_0))$$

$$+ \frac{\partial \hat{\chi}^{(3)}}{\partial \omega_2}(x, -\omega_0, \omega_0, \omega_0)(\omega_2 - \omega_0) + \frac{\partial \hat{\chi}^{(3)}}{\partial \omega_3}(x, -\omega_0, \omega_0, \omega_0)(\omega_3 - \omega_0 + \dots)$$

$$\times (\hat{A}^*(Z, -\omega_1 - \omega_0) \hat{A}(Z, \omega_2 - \omega_0) \hat{A}(Z, \omega_3 - \omega_0) U^*(x, \omega_0) \exp(ik_0 z)) \quad (3.25)$$

where we assumed U to be real and even in ω ($U^*(x, \omega_0) = U(x, \omega_0) = U(x, \omega_0)$). Taking

the inverse Fourier transform of (3.24), and replacing $\delta(\omega - \omega_1 - \omega_2 - \omega_3)$ with the

integration over t' in the same way as (3.22), we have

$$-\frac{1}{(2\pi)^4} \int d\omega \exp(i\omega t) \iiint d\omega_1 d\omega_2 d\omega_3 \int dt' \exp[i(\omega - \omega_1 - \omega_2 - \omega_3)t']$$

$$\times \hat{A}^*(Z, -\omega_1 - \omega_0) \hat{A}(Z, \omega_2 - \omega_0) \hat{A}(Z, \omega_3 - \omega_0) U^3(x, \omega_0) \exp(ik_0 z)$$

$$\times (\omega_0^2 \hat{\chi}^{(3)}(x, -\omega_0, \omega_0, \omega_0) + \omega_0^2 \frac{\partial \hat{\chi}^{(3)}}{\partial \omega_1}(x, -\omega_0, \omega_0, \omega_0)(\omega_1 + \omega_0)$$

$$+ \omega_0^2 \frac{\partial \hat{\chi}^{(3)}}{\partial \omega_2}(x, -\omega_0, \omega_0, \omega_0)(\omega_2 + \omega_0) + \omega_0^2 \frac{\partial \hat{\chi}^{(3)}}{\partial \omega_3}(x, -\omega_0, \omega_0, \omega_0)(\omega_3 + \omega_0) + \dots)$$

$$\begin{aligned}
&= -\frac{1}{(2\pi)^4} \iiint d\omega d\omega_1 d\omega_2 d\omega_3 dt' \exp[-i\omega(t-t')] \exp[-i(\omega_1 + \omega_2 + \omega_3)t'] \\
&\times [\omega_0^2 \hat{\chi}^{(3)}(x, -\omega_0, \omega_0, \omega_0) + \dots] U^3(x, \omega_0) \hat{A}^*(Z, -\omega_1 - \omega_0) \hat{A}(Z, -\omega_2 - \omega_0) \hat{A}(Z, -\omega_3 - \omega_0) \quad (3.26) \\
&\times \exp(ik_0 z).
\end{aligned}$$

We note that the integration over ω in (3.25) can be written as

$$\int_{-\infty}^{\infty} \exp[-i\omega(t-t')] d\omega = 2\pi\delta(t-t'), \quad (3.27)$$

and thus the integration over t' is equivalent to simply replacing t' with t . It is also

convenient to write

$$\begin{aligned}
&\exp[-i(\omega_1 + \omega_2 + \omega_3)t'] \\
&= \exp(i\omega_0 t') \exp(-i(\omega_1 + \omega_0)t') \exp(-i(\omega_2 - \omega_0)t') \exp(-i(\omega_3 - \omega_0)t') \\
&\equiv \exp(i\omega_0 t') \exp(-i(\omega_1 + \omega_0)t') \exp(-i(\omega_2 - \omega_0)t') \exp(-i(\omega_3 - \omega_0)t') \\
&\mathfrak{E}\Omega_1 \equiv \omega_1 + \omega_0, \mathfrak{E}\Omega_2 \equiv \omega_2 + \omega_0, \mathfrak{E}\Omega_3 \equiv \omega_3 + \omega_0. \quad (3.28)
\end{aligned}$$

Consequently (3.25) becomes, to leading order,

$$\begin{aligned}
&-\frac{\varepsilon^3}{(2\pi)^2} \iiint d\Omega_1 d\Omega_2 d\Omega_3 \\
&\times \exp(i\omega_0 t) (\exp(-i\Omega_1 T) \hat{A}^*(Z, -\mathfrak{E}\Omega_1)) (\exp(-i\Omega_2 T) \hat{A}(Z, \mathfrak{E}\Omega_2)) \\
&\times (\exp(-i\Omega_3 T) \hat{A}(Z, \mathfrak{E}\Omega_3)) \omega_0^2 \hat{\chi}^{(3)}(x, -\omega_0, \omega_0, \omega_0) U^3(x, \omega_0) \exp(ik_0 z) \\
&= -\frac{\varepsilon^3}{(2\pi)^2} \exp(ik_0 z - i\omega_0 t) \omega_0^2 \hat{\chi}^{(3)}(x, -\omega_0, \omega_0, \omega_0) U^3(x, \omega_0) \\
&\times \int d\Omega_1 \hat{A}^*(Z, -\mathfrak{E}\Omega_1) \exp(-i\Omega_1 T) \int d\Omega_2 \hat{A}(Z, \mathfrak{E}\Omega_2) \exp(-i\Omega_2 T) \\
&\times \int d\Omega_3 \hat{A}(Z, \mathfrak{E}\Omega_3) \exp(-i\Omega_3 T) \\
&= -\varepsilon^3 \omega_0^2 \hat{\chi}^{(3)}(x, -\omega_0, \omega_0, \omega_0) A^2(Z, T; \varepsilon) U^3(x, \omega_0) \exp(ik_0 z - i\omega_0 t). \quad (3.29)
\end{aligned}$$

The extension of the calculation to other nonlinear terms in (3.21) proportional to

$\exp(\pm ik_0 z)$ is straightforward. We finally obtain the nonlinear polarization of the form

$$\begin{aligned}
& -3\varepsilon^3 \omega_0^2 \hat{\chi}^{(3)}(x, -\omega_0, \omega_0, \omega_0) A^2(Z, T) A^*(Z, T) U^3(x, \omega_0) \exp(ik_0 z - i\omega_0 t) \\
& -3\varepsilon^3 \omega_0^2 \hat{\chi}^{(3)}(x, -\omega_0, \omega_0, \omega_0) A^{*2}(Z, T) A(Z, T) U^3(x, \omega_0) \exp(-ik_0 z + i\omega_0 t)
\end{aligned} \tag{3.30}$$

as a result of the quasi-monochromatic approximation.

3.4 Asymptotic expansion

With these preparations, we now return to the Maxwell's equations (3.16) and apply the multiple scale method to derive an approximate equation that describes the long range evolution of the slowly varying wave envelope. We asymptotically expand the electric field E as (3.17), and introduce multiple scales with respect to z :

$$Z_1 = \varepsilon z, \quad Z_2 = \varepsilon^2 z, \quad \dots, \quad Z_n = \varepsilon^n z. \tag{3.31}$$

Note that we do not observe the role of the nonlinear polarization until we proceed to $O(\varepsilon^3)$. At the lowest order $O(\varepsilon)$ we expect (3.16) simply gives the guiding mode of optical wave packets. At the second lowest order $O(\varepsilon^2)$, where the nonlinearity is still out of scope, (3.16) is expected to give rise to the long term behavior of the guided wave propagation along z with the group velocity of $1/k'_0$ where $k'_0 \equiv \partial k / \partial \omega|_{\omega=\omega_0}$. It is convenient to introduce the linear operator L to represent the linear response of the electric field:

$$L\left(\frac{d}{dx}, ik, -i\omega\right) \equiv \frac{d^2}{dx^2} - k^2 + \frac{n^2(x, \omega)\omega^2}{c^2}, \tag{3.32}$$

where $n(x, \omega)$ and c represent the refractive index of silica and the velocity of light for free space, respectively. As far as the linear response has no retardation, L is written as

$$L\left(\frac{d}{dx}, ik, -i\omega\right) \equiv \frac{d^2}{dx^2} - k^2 + \frac{n_0^2 \omega_0^2}{c^2}, \tag{3.33}$$

where $n_0 \equiv n(x, \omega_0)$. With the operator L , the linear component of (3.16) has the form

$$\begin{aligned} & \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \int d\tau \chi^{(1)}(x, t - \tau) \right) (E_0 + \varepsilon E^1 + \dots) \\ &= L \left(\frac{d}{dx}, ik_0 + \varepsilon \frac{\partial}{\partial z_1} + \varepsilon^2 \frac{\partial}{\partial z_2} + \dots, -i\omega_0 + \varepsilon \frac{\partial}{\partial T} \right) \\ & \times [\varepsilon A(Z_1, Z_2, \dots, T) U(x, \omega_0) \exp(ik_0 z - i\omega_0 t) + c.c.], \end{aligned} \quad (3.34)$$

where we referred to (3.19). Note that the dispersion relation $k = k(\omega)$, obtained as the eigenvalue in solving the boundary value problem of the form $LV=0$ (where

$E = \varepsilon V \exp(ik_0 z - i\omega_0 t) + c.c.$, $V \equiv U(x, \omega_0) A(Z, T)$) is given by the relation

$$D([V], k, \omega) \equiv \langle V, LV \rangle = 0, \quad (3.35)$$

where the brackets denote the inner product:

$$\langle A, B \rangle \equiv \int A^* B dx. \quad (3.36)$$

Taking the integration by parts, we obtain

$$\begin{aligned} k^2 &= \frac{\frac{\omega^2}{c^2} \int n^2(x, \omega) |V|^2 dx - \int \left| \frac{\partial V}{\partial x} \right|^2 dx}{\int |V|^2 dx} \\ &= \frac{\frac{\omega^2}{c^2} \int n^2(x, \omega) U^2(x, \omega_0) dx - \int \left(\frac{dU}{dx}(x, \omega_0) \right)}{\int U^2(x, \omega_0) dx} \end{aligned} \quad (3.37)$$

$O(\varepsilon)$: the transverse structure

Extracting $O(\varepsilon)$ contributions from (3.16), we have

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \int d\tau \chi^{(1)}(x, t - \tau) E_0 = 0. \quad (3.38)$$

With (3.33), we obtain

$$L\left(\frac{d}{dx}, ik_0, -i\omega_0\right)AU(x, \omega_0)\exp(ik_0z - i\omega_0t) + c.c. = 0. \quad (3.39)$$

We set E_0 as following form

$$E_0 = \varepsilon U_0 \exp(ik_0z - i\omega_0t) + c.c. \quad (3.40)$$

and find

$$L\left(\frac{d}{dx}, ik_0 - i\omega_0\right)U_0 = \frac{d^2U_0}{dx^2} - k_0^2U_0 + \frac{n_0^2\omega_0^2}{c^2}U_0 = 0, \quad (3.41)$$

which simply gives the transverse structure of E_0 .

$O(\varepsilon^2)$: linear propagation of wave packets

Before studying the $O(\varepsilon^2)$ contribution, we expand L in a Taylor series in advance:

$$\begin{aligned} L &= L\left(\frac{d}{dx}, ik_0 + \varepsilon \frac{\partial}{\partial Z_1} + \varepsilon^2 \frac{\partial}{\partial Z_2} + \dots, -i\omega_0 + \varepsilon \frac{\partial}{\partial T}\right) \\ &\equiv L(X, K, \Omega) \\ &= L_1 \\ &\quad + \frac{\partial L}{\partial K}\bigg|_{K_0} (K - K_0) + \frac{\partial L}{\partial \Omega}\bigg|_{\Omega_0} (\Omega - \Omega_0) \\ &\quad + \frac{1}{2} \frac{\partial^2 L}{\partial K^2}\bigg|_{K_0} (K - K_0)^2 + \frac{1}{2} \frac{\partial^2 L}{\partial \Omega^2}\bigg|_{\Omega_0} (\Omega - \Omega_0)^2 + \frac{1}{2} \frac{\partial^2 L}{\partial K^2 \partial \Omega^2}\bigg|_{K_0, \Omega_0} (K - K_0)(\Omega - \Omega_0) \\ &= L_1 \\ &\quad + L_2(\varepsilon \frac{\partial^2 \Omega}{\partial u^2}) + L_3(\varepsilon \frac{\partial^2 \Omega}{\partial u^2}) \\ &\quad + L_2(\varepsilon \frac{\partial}{\partial Z_2}) + \frac{1}{2} L_{22}(\varepsilon^2 \frac{\partial^2}{\partial Z_1^2}) + \frac{1}{2} L_{33}(\varepsilon^2 \frac{\partial^2}{\partial T^2}) + \frac{1}{2} L_{23}(\varepsilon^2 \frac{\partial^2}{\partial Z_1 \partial T}) \\ &\quad + \dots, \end{aligned} \quad (3.42)$$

where we define the following parameters for convenience

$$\begin{aligned}
X &\equiv \frac{d}{dx} \\
K - K_0 &\equiv ik - ik_0 = \varepsilon \frac{\partial}{\partial Z_1} + \varepsilon^2 \frac{\partial}{\partial Z_2} + \dots, \\
\Omega - \Omega_0 &\equiv -i\omega - (-i\omega_0) = \varepsilon \frac{\partial}{\partial T}, \\
L_1 &\equiv L(X, K_0 \Omega_0),
\end{aligned} \tag{3.43}$$

$$L_2 \equiv \frac{\partial L}{\partial K} \Big|_{K_0}, L_2 \equiv \frac{\partial L}{\partial \Omega} \Big|_{\Omega_0}, L_2 \equiv \frac{\partial^2 L}{\partial K^2} \Big|_{K_0}, \dots \tag{3.44}$$

We also calculate the differentiation of the dispersion relation $D([V], k, \omega) = \langle V, LV \rangle = 0$ with respect to ω , which is used to obtain relations between certain inner products.

Since (3.34) holds for all ω , we have

$$\frac{dD}{d\omega} = \left\langle \frac{dV}{d\omega}, LV \right\rangle + \left\langle V, L \frac{dV}{d\omega} \right\rangle + \left\langle V, \frac{dL}{d\omega} V \right\rangle = 0 \tag{3.45}$$

Note in (3.43) that $\langle V, LX \rangle = \langle X, LV \rangle$ for L being a self-adjoint operator. Thus we obtain

$$2 \left\langle \frac{dV}{d\omega}, LV \right\rangle + \left\langle V, \frac{dL}{d\omega} V \right\rangle = 0 \tag{3.46}$$

Recall that on a solution U at $O(\varepsilon)$, $L_1 U = 0$ in (3.44), so that $k = k_0$ and

$$\begin{aligned}
\left\langle U, \frac{dL}{d\omega} \Big|_{k_0, \omega_0} U \right\rangle &= \left\langle U, (L_2 \frac{d(ik)}{d\omega} \Big|_{\omega_0} + L_3 \frac{d(-i\omega)}{d\omega} \Big|_{\omega_0}) U \right\rangle \\
\omega = \omega_0 &= \left\langle U, (ik'_0 L_2 - iL_3) U \right\rangle \\
&= ik'_0 \langle U, L_2 U \rangle - i \langle U, L_3 U \rangle \\
&= 0.
\end{aligned} \tag{3.47}$$

At $O(\varepsilon^2)$, we have from (3.16)

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \int d\tau \chi^{(1)}(x, t - \tau)\right) \varepsilon E_1 = 0. \quad (3.48)$$

With (3.41), we obtain the following

$$\text{relation } L_1(\varepsilon E_1) + \varepsilon^2 (L_2 U(x, \omega_0) \frac{\partial A}{\partial Z_1} + L_3 U(x, \omega_0) \frac{\partial A}{\partial T}) \exp(ik_0 z - i\omega_0 t) + c.c. = 0. \quad (3.49)$$

We put $E_1 = \varepsilon U_1 \exp(ik_0 z - i\omega_0 t) + c.c.$, so that we have

$$L_1 U_1 = F_1 \equiv -(L_2 U(x, \omega_0) \frac{\partial A}{\partial Z_1} + L_3 U(x, \omega_0) \frac{\partial A}{\partial T}) \quad (3.50)$$

Solvability condition of E_1 requires that the transverse component $U(x)$ is a smooth solution that vanishes at the boundary $|x| \rightarrow \infty$. In mathematical literature, the requirement is described as the Fredholm alternative theorem; the inhomogeneous boundary value problem $\Lambda X = F$, with the singular operator Λ having nontrivial homogeneous solutions V such that $\Lambda V = 0$, has the bound state solution when the right hand side F is orthogonal to V :

$$\langle V, F \rangle = \langle V, \Lambda X \rangle = 0. \quad (3.51)$$

The brackets denote the inner product defined by (3.35). For the problem (3.48), the Fredholm theorem demands thereby

$$\langle U, F_1 \rangle = \langle U, -L_2 U \rangle \frac{\partial A}{\partial Z_1} + \langle U, -L_3 U \rangle \frac{\partial A}{\partial T} = 0. \quad (3.52)$$

From (3.45), we have

$$\langle U, -L_2 U \rangle \left(\frac{\partial A}{\partial Z_1} + K_0 \frac{\partial A}{\partial T} \right) = 0. \quad (3.53)$$

Note that

$$\langle U, -L_2 U \rangle = 2ik_0 \langle U, U \rangle \neq 0, \quad (3.54)$$

consequently the solvability condition leads to the following equation

$$\frac{\partial A}{\partial Z_1} + K_0' \frac{\partial A}{\partial T} = 0. \quad (3.55)$$

which describes the propagation of a wave packet at the group velocity of $1/k_0'$. The solvability condition gives rise to the slow variation of the mode distribution $U(x)$ with respect to ω . This can be clearly seen by evaluating the forcing function F_1 namely

$$\begin{aligned} F_1 &= -(L_2 U \frac{\partial A}{\partial Z_1} + L_3 U \frac{\partial A}{\partial T}) \\ &= (L_2 k_0' - L_3) U \frac{\partial A}{\partial T} \\ &= (-iL_2 k_0' - iL_3) U (i \frac{\partial A}{\partial T}) \\ &= -(\frac{\partial L}{\partial K} \frac{dK}{d\omega_0} \frac{\partial A}{\partial Z_1} + \frac{\partial L}{\partial \omega_0}) U (i \frac{\partial A}{\partial T}) \\ &= -i \frac{dL}{\omega_0} U \frac{\partial A}{\partial T}, \end{aligned} \quad (3.56)$$

where we used (3.53) and $d/d\omega_0 \equiv d/d\omega|_{\omega_0}$. On the other hand, differentiation of

$L_1 U = 0$ with respect to ω leads to $dL/d\omega_0 = 0$ and thus $F_1 = 0$, unless U has ω

dependency. Therefore (3.54) can now be written as

$$F_1 = -iL \frac{\partial U}{\partial \omega_0} \frac{\partial A}{\partial T}, \quad (3.57)$$

This gives rise to nonzero U_1 :

$$U_1 = -i \frac{\partial U}{\partial \omega_0} \frac{\partial A}{\partial T}. \quad (3.58)$$

Thus we have at $O(\varepsilon^2)$

$$E_0 + \varepsilon E_1 = \varepsilon(U(x, \omega_0) + i\varepsilon \frac{\partial U}{\partial \omega}(x, \omega_0) \frac{\partial}{\partial T})A(Z, T)\exp(ik_0 z - i\omega_0 t) + c.c. \quad (3.59)$$

$O(\varepsilon^3)$: the nonlinear Schrödinger equation

Before proceeding to the next order, we make several preparations again. We

differentiate (3.43) again with respect to ω . Using (3.45), we find

$$\frac{d^2 D}{d\omega^2} = \left\langle \frac{dV}{d\omega}, \frac{dL}{d\omega} V \right\rangle + \left\langle V, \frac{dL}{d\omega} \frac{dV}{d\omega} \right\rangle + \left\langle V, \frac{d^2 L}{d\omega^2} V \right\rangle = 2 \left\langle \frac{dL}{d\omega} \frac{dV}{d\omega} \right\rangle + \left\langle V, \frac{d^2 L}{d\omega^2} V \right\rangle = 0, \quad (3.60)$$

which gives (on a solution U for $k = k_0$ and $\omega = \omega_0$)

$$2 \left\langle U, (ik'_0 L_2 - iL_3 \frac{\partial U}{\partial \omega_0}) \right\rangle + ik'_0 \langle U, L_2 U \rangle - k_0'^2 \langle U, L_{22} U \rangle + 2k'_0 \langle U, L_{23} U \rangle - \langle U, L_{33} U \rangle = 0, \quad (3.61)$$

where we used

$$\frac{dL}{d\omega_0} = ik'_0 L_2 - iL_3, \quad \frac{dL}{d\omega_0} = ik'_0 L_2 - k_0'^2 L_{22} + 2k'_0 L_{23} - L_{33}. \quad (3.62)$$

At $O(\varepsilon^3)$, we have to take into consideration the nonlinear polarization. Extracting the

$O(\varepsilon^2)$ contributions from (3.16), we obtain

$$\begin{aligned} & \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \int d\tau \chi^{(1)}(x, t - \tau) \varepsilon^2 E_2 \\ & - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \iiint \chi^{(3)}(x, t - \tau_1, t - \tau_2, t - \tau_3) E_0(x, z, \tau_1) E_0(x, z, \tau_2) E_0(x, z, \tau_3) d\tau_1 d\tau_2 d\tau_3 = 0. \end{aligned} \quad (3.63)$$

E_2 has the phase components $\exp(\pm ik_0 z \mp i\omega_0 t)$ as well as $\exp(\pm 3ik_0 z \mp 3i\omega_0 t)$, which we

explicitly represent as

$$E_2 = \varepsilon U_2 \exp(ik_0 z - i\omega_0 t) + \varepsilon^3 V_2 \exp(3ik_0 z - 3i\omega_0 t) + c.c. \quad (3.64)$$

The nonlinear polarization term contains the component proportional to $\exp(ik_0 z - i\omega_0 t)$ given by

$$-\varepsilon^3 \frac{3\omega_0^2}{c^2} \hat{\chi}^{(3)}(x, -\omega_0, \omega_0, \omega_0) A^2(Z, T) A^*(Z, T) U^3(x, \omega_0) \exp(ik_0 z - i\omega_0 t), \quad (3.65)$$

where we referred to (3.29). As a consequence, using the linear operator L , (3.61) can be written as the following arranged form:

$$L(\varepsilon^2 E_2) = \varepsilon^3 F_2 \exp(ik_0 z - i\omega_0 t) + \varepsilon^3 G_2 (3ik_0 z - 3i\omega_0 t) + c.c. \quad (3.66)$$

namely, for the frequency ω_0 and the wave number k_0 we have

$$L_1 U_2 = F_2,$$

where

$$\begin{aligned} F_2 = & -L_2 U \frac{\partial A}{\partial Z_2} - \frac{1}{2} L_{22} U \frac{\partial^2 A}{\partial Z_1^2} - \frac{1}{2} L_{33} U \frac{\partial^2 A}{\partial T^2} - L_{23} U \frac{\partial^2 A}{\partial Z_1 \partial T} \\ & - iL_2 \frac{\partial U}{\partial \omega_0} \frac{\partial^2 A}{\partial Z_1 \partial T} - iL_3 \frac{\partial U}{\partial \omega_0} \frac{\partial^2 A}{\partial T^2} \\ & - \frac{3\omega_0^2}{c^2} \hat{\chi}^{(3)}(x, -\omega_0, \omega_0, \omega_0) A^2(Z, T) A^*(Z, T) U^3(x, \omega_0). \end{aligned} \quad (3.67)$$

In terms of the linear electric polarization, the first line of (3.66) originates from εE_0 and the $O(\varepsilon^2)$ contributions of L , whereas the second line is obtained from $\varepsilon^2 E_1$ and the $O(\varepsilon)$ contributions of L . (See (3.41) and (3.57)). Following the analysis at $O(\varepsilon^2)$, we apply the Fredholm's theorem to (3.65):

$$\langle U, F_2 \rangle = 0, \quad (3.68)$$

where the left hand side of (3.67) can be written as

$$\begin{aligned}
\langle U, F_2 \rangle &= \langle U, -L_2 U \rangle \frac{\partial A}{\partial Z_2} + \frac{1}{2} \langle U, -L_{22} U \rangle \frac{\partial^2 A}{\partial Z_1^2} + \frac{1}{2} \langle U, -L_{33} U \rangle \frac{\partial^2 A}{\partial T^2} + \frac{1}{2} \langle U, -L_{23} U \rangle \frac{\partial^2 A}{\partial Z_1 \partial T} \\
&+ i \left\langle U, -L_2 \frac{\partial U}{\partial \omega_0} \right\rangle \frac{\partial^2 A}{\partial Z_1 \partial T} + i \left\langle U, -L_3 \frac{\partial U}{\partial \omega_0} \right\rangle \frac{\partial^2 A}{\partial T^2} - \frac{3\omega_0^2}{c^2} \langle U, \hat{\chi}^{(3)} U^3 \rangle A^2 A^* \\
&= \langle U, -L_2 U \rangle \frac{\partial A}{\partial Z_2} + \left(\frac{1}{2} k_0' \langle U, L_{22} U \rangle - \langle U, L_{23} U \rangle \right) \frac{\partial^2 A}{\partial Z_1 \partial T} + \frac{1}{2} \langle U, -L_{33} U \rangle \frac{\partial^2 A}{\partial T^2} \\
&+ (-i \left\langle U, L_2 \frac{\partial U}{\partial \omega_0} \right\rangle + \frac{i}{k_0'} \left\langle U, -L_3 \frac{\partial U}{\partial \omega_0} \right\rangle) \frac{\partial^2 A}{\partial Z_1 \partial T} - \frac{3\omega_0^2}{c^2} \langle U, \hat{\chi}^{(3)} U^3 \rangle A^2 A^* \\
&= \langle U, -L_2 U \rangle \frac{\partial A}{\partial Z_2} - i \frac{k_0'}{2k_0'} \langle U, -L_2 U \rangle \frac{\partial^2 A}{\partial Z_1 \partial T} + \frac{1}{2k_0'} \langle U, -L_{33} U \rangle \\
&\times \left(\frac{\partial^2 A}{\partial Z_1 \partial T} + k_0' \frac{\partial^2 A}{\partial T^2} \right) - \frac{3\omega_0^2}{c^2} \langle U, \hat{\chi}^{(3)} U^3 \rangle A^2 A^* \\
&= \langle U, -L_2 U \rangle \left(\frac{\partial A}{\partial Z_2} + i \frac{k_0'}{2k_0'} \frac{\partial^2 A}{\partial T^2} \right) - \frac{3\omega_0^2}{c^2} \langle U, \hat{\chi}^{(3)} U^3 \rangle A^2 A^* \\
&= -2ik_0 \langle U, U \rangle \left(\frac{\partial A}{\partial Z_2} + i \frac{k_0'}{2k_0'} \frac{\partial^2 A}{\partial T^2} \right) - \frac{3\omega_0^2}{c^2} \langle U, \hat{\chi}^{(3)} U^3 \rangle A^2 A^* \\
&= -2ik_0 \int U^2(x, \omega_0) dx \left(\frac{\partial A}{\partial Z_2} + i \frac{k_0'}{2k_0'} \frac{\partial^2 A}{\partial T^2} \right) \\
&- \frac{3\omega_0^2}{c^2} A^2 A^* \int \hat{\chi}^{(3)}(x, -\omega_0, \omega_0, \omega_0) U^4(x, \omega_0) dx.
\end{aligned} \tag{3.69}$$

We have referred to (3.59) and the following relations

$$\frac{\partial^2 A}{\partial Z_1 \partial T} + k_0' \frac{\partial^2 A}{2T^2} = 0, \quad \frac{\partial^2 A}{\partial Z_1^2} + k_0' \frac{\partial^2 A}{\partial Z_1 \partial T} = 0, \tag{3.70}$$

which are obtained by the differentiation of (3.53) with respect to T and Z_1 respectively.

We now have

$$-2ik_0 \int U^2(x, \omega_0) dx \left(\frac{\partial A}{\partial Z_2} + i \frac{k_0'}{2} \frac{\partial^2 A}{\partial T^2} \right) - \frac{3\omega_0^2}{c^2} A^2 A^* \int \hat{\chi}^{(3)}(x, -\omega_0, \omega_0, \omega_0) U^4(x, \omega_0) dx = 0 \tag{3.71}$$

or simply,

$$\frac{\partial A}{\partial Z_2} + i \frac{k_0''}{2} \frac{\partial^2 A}{\partial T^2} + \frac{-\frac{3\omega_0^2}{c^2} \int \hat{\chi}^{(3)} U^4 dx}{-2k_0 \int U^2 dx} A^2 A^* = 0 \quad (3.72)$$

We hereafter assume for simplicity the nonlinear susceptibility $\hat{\chi}^{(3)}$ having negligibly small dependence on the transverse coordinate x . We define the following parameters

$$n_0 \equiv \frac{ck_0}{\omega_0}, \quad n_2 \equiv \frac{3\hat{\chi}^{(3)}(\omega_0)}{2n_0}, \quad g \equiv \frac{\int U^4 dx}{\int U^2 dx}, \quad (3.73)$$

so that we have

$$\frac{\partial A}{\partial Z_2} + i \frac{k_0'}{2} \frac{\partial^2 A}{\partial T^2} - i \frac{gn_2\omega_0}{c} A^2 A^* = 0. \quad (3.74)$$

Note that in the definition in (3.72), n_0 and n_2 corresponds to the linear and the third order nonlinear coefficients of the refractive index:

$$n(\omega_0, |E|) = n_0 + n_2 |E|^2. \quad (3.75)$$

Combining (3.53) and (3.73), and introducing the new coordinates

$$\tau = \varepsilon(t - k_0'), \quad \zeta = \varepsilon^2 z, \quad (3.76)$$

we finally obtain the canonical form of the nonlinear Schrödinger equation

$$\frac{\partial A}{\partial \zeta} + i \frac{k_0'}{2} \frac{\partial^2 A}{\partial \tau^2} - i \frac{gn_2\omega_0}{c} A^2 A^* = 0 \quad (3.77)$$

which in the dimensionless form reduces to,

$$iq_t + \frac{1}{2} q_{xx} + |q|^2 q = 0 \quad (3.78)$$

after change of variables.

CHAPTER IV

SOLUTION FOR A PERTURBED NON-LINEAR SCHRÖDINGER'S EQUATION

In this chapter, the nonlinear Schrödinger's equation (NLSE)

$$iq_t + \frac{1}{2}q_{xx} + |q|^2 q = iR \quad (4.1)$$

where $R = \nu |q|^2 q_x - i\sigma(|q|^2)_x q$, where the coefficient of ν , a perturbation term, called nonlinear dispersion that arises in nonlinear optics will be considered and σ is called the Raman scattering.

Therefore the equation of study becomes

$$iq_t + \frac{1}{2}q_{xx} + |q|^2 q = i\nu |q|^2 q_x + \sigma(|q|^2)_x q \quad (4.2)$$

4.1 Traveling Waves

Let's seek waves of permanent form by trying the solution ansatz

$$q = re^{i(\theta + nt)} \quad (4.3)$$

Where r and θ are real functions of $x - ct$ for some real constants n and c .

$$\text{That is } \theta = \theta(x - ct), \text{ and } r = r(x - ct). \quad (4.4)$$

where $c, n \in \mathbb{R}$

This problem above can be solved by finding an expression for each term in the given equation with the help of (4.2) and then substitute all the expressions into (4.1)

Therefore differentiating q with respect to t , we have

$$\begin{aligned} q_t &= -cr' e^{i(\theta+nt)} + r e^{i(\theta+nt)} (i\theta' c + in) \\ &= e^{i(\theta+nt)} [-cr' + r(in - ic\theta')] \end{aligned} \quad (4.5)$$

$$iq_t = e^{i(\theta+nt)} (-icr' - nr + cr\theta') \quad (4.6)$$

Subsequently, finding the first derivative of q with respect to x , we have

$$q_x = r' e^{i(\theta+nt)} + r e^{i(\theta+nt)} (i\theta') \quad (4.7)$$

$$q_x = e^{i(\theta+nt)} (r' + ir\theta') \quad (4.8)$$

Also finding the second derivative of U with respect to x , we have

$$\begin{aligned} q_{xx} &= e^{i(\theta+nt)} (r'' + ir\theta'' + ir'\theta') + (r' + ir\theta') e^{i(\theta+nt)} (i\theta') \\ q_{xx} &= e^{i(\theta+nt)} [r'' + ir\theta'' + 2ir'\theta' - r(\theta')^2] \end{aligned} \quad (4.9)$$

Simplifying and multiplying through by $\frac{1}{2}$, we have

$$\frac{1}{2} q_{xx} = \frac{1}{2} e^{i(\theta+nt)} [(r'' + ir\theta'' + 2ir'\theta' - r(\theta')^2)] \quad (4.10)$$

Solving for the term $|q|^2 = q \cdot q^*$ where q^* is the conjugate of q

We have $|q|^2 = q \cdot q^* = r e^{i(\theta+nt)} * r e^{-i(\theta+nt)} = r^2$

Hence $|q|^2 q = r^2 * r e^{i(\theta+nt)} = r^3 e^{i(\theta+nt)}$ (4.11)

Finally $v |q|^2 q_x = v r^2 e^{i(\theta+nt)} (r' + ir\theta')$

$$iv |q|^2 q_x = e^{i(\theta+nt)} (iv r^2 r' - v r^3 \theta') \quad (4.12)$$

$$\sigma(|q|^2)_x q = \sigma e^{i(\theta+nt)} (2r^2 r')$$

Hence substituting (4.4), (4.8), (4.9) and (4.10) into (4.1), we have

$$-icr' - nr + cr\theta' + \frac{r''}{2} + i\frac{r\theta''}{2} + ir'\theta' - \frac{r(\theta')^2}{2} + r^3 - ivr^2r' + vr^3\theta' - 2\sigma r^2r' = 0 \quad (4.13)$$

Now, simplifying (4.11) by grouping all imaginary components and real components and equating them to zero we have

Imaginary part:

$$-cr' + \frac{r\theta''}{2} + r'\theta' - vr^2r' = 0 \quad (4.14)$$

Real part:

$$-nr + cr\theta' + \frac{r''}{2} + -\frac{r(\theta')^2}{2} + r^3 + vr^3\theta' - 2\sigma r^2r' = 0 \quad (4.15)$$

Now, considering (4.12), multiply both sides of the equation

by $2r$ and rearranging, we have

$$r^2\theta'' + 2rr'\theta' - 2crr' - 2vr^3r' = 0 \quad (4.16)$$

Simplifying (4.14), we have

$$(r^2\theta')' - c(r^2)' - \frac{v}{2}(r^4)' = 0 \quad (4.17)$$

Hence integrating (4.15), we have

$$(r^2\theta') - c(r^2) - \frac{v}{2}(r^4) = A \quad (4.18)$$

where A is the integration constant

simplifying (4.16) we have

$$\theta' = \left(c + \frac{vs}{2} + \frac{A}{s}\right), \quad (4.19)$$

$$\text{where } s = r^2 \quad (4.20)$$

$$\text{Now } (s')^2 = -2G(s), \quad (4.21)$$

where $G(s)$ is the function of a given wave of permanent form

Substituting (4.18) into (4.19) and simplifying we have

$$[(r^2)']^2 = -2G(s) \quad (4.22)$$

differentiating (4.20) we have

$$(2rr')^2 = -2G(s) \quad (4.23)$$

simplifying and substituting (4.21) we have

$$4s(r')^2 = -2G(s) \quad (4.24)$$

Now, to be able to find the solutions, to the given cnoidal wave, which is a function expressed in $G(s)$, we have to first substitute (4.17) into (4.13) and solve for $F(s)$.

Therefore substituting (4.17) into (4.13) we have

$$\frac{r''}{2} + vr^3(c + \frac{vs}{2} + \frac{A}{s}) - \frac{r}{2}(c + \frac{vs}{2} + \frac{A}{s})^2 + r^3 + cr(c + \frac{vs}{2} + \frac{A}{s}) - nr - 2\sigma r^2 r' = 0 \quad (4.25)$$

Expanding and opening the parentheses, we have

$$\begin{aligned} & \frac{r''}{2} + \frac{vr^3 A}{s} + vr^3 c + \frac{v^2 r^3 s}{2} - \frac{rc^2}{2} - \frac{cAr}{s} - \frac{rvsc}{2} - \frac{rvA}{2} \\ & - \frac{rA^2}{s^2} - \frac{v^2 rs^2}{4} + r^3 + \frac{cAr}{s} + c^2 r + \frac{crvs}{2} - 2\sigma r^2 r' - nr = 0 \end{aligned} \quad (4.26)$$

Simplifying by substituting $(s = r^2)$ and grouping like terms of (4.24), we have

$$\frac{r''}{2} - (n + \frac{vA}{4} + \frac{c^2}{2})r - \frac{A^2}{r^3} + (vc + 1)r^3 + \frac{v}{4}r^5 - 2\sigma r^2 r' = 0 \quad (4.27)$$

Multiplying through (4.25) by r' , we have

$$\frac{r''}{2}r' - \left(n + \frac{\nu A}{4} + \frac{c^2}{2}\right)rr' - \frac{A^2}{r^3}r' + (\nu c + 1)r^3r' + \frac{\nu}{4}r^5r' - 2\sigma r^2(r')^2 = 0 \quad (4.28)$$

Simplifying, we have

$$\left[\frac{(r')^2}{4}\right]' - \left(n + \frac{\nu A}{4} + \frac{c^2}{2}\right)\left(\frac{r^2}{2}\right)' - \left(\frac{A^2}{2r^2}\right)' + (\nu c + 1)\left(\frac{r^4}{4}\right)' + \frac{\nu}{4}\left(\frac{r^6}{6}\right)' - 2\sigma \int r^2(r')^2 dr = 0 \quad (4.29)$$

Integrating (4.27), we have

$$\frac{(r')^2}{4} - \left(n + \frac{\nu A}{4} + \frac{c^2}{2}\right)\left(\frac{r^2}{2}\right) - \left(\frac{A^2}{2r^2}\right) + (\nu c + 1)\left(\frac{r^4}{4}\right) + \frac{\nu}{4}\left(\frac{r^6}{6}\right) - 2\sigma \int r^2(r')^2 dr = B \quad (4.30)$$

where B is integration constant.

Substituting back $s = r^2$ into (4.28) and simplifying, then

$$(s')^2\left(\frac{1}{4} + \sigma\right) = \left(n + \frac{\nu A}{4} + \frac{c^2}{2}\right)\left(\frac{s}{2}\right) + \left(\frac{A^2}{2s}\right) - (\nu c + 1)\left(\frac{s^2}{4}\right) - \frac{\nu}{4}\left(\frac{s^3}{6}\right) + B \quad (4.31)$$

Multiply through (4.29) by $24s$; we have

$$(s')^2(6s + 24s\sigma) = \left(n + \frac{\nu A}{4} + \frac{c^2}{2}\right)(12s^2) + (12A^2) - (\nu c + 1)(6s^3) - \nu s^4 + 24Bs \quad (4.32)$$

Simplifying (4.30) further and rearranging, we have

$$(s')^2 = \frac{\left(n + \frac{\nu A}{4} + \frac{c^2}{2}\right)(12s^2) + (12A^2) - (\nu c + 1)(6s^3) - \nu s^4 + 24Bs}{(6s + 24s\sigma)} \quad (4.33)$$

which can be reduced to,

$$(s')^2 = B_1 + B_2s + B_3s^2 + B_4s^3 \quad (4.34)$$

where $B_i = B_i(A, B, n, \nu, \sigma)$

for $1 \leq i \leq 4$

$$\Rightarrow \left(\frac{ds}{d\xi}\right) = \sqrt{B_1 + B_2s + B_3s^2 + B_4s^3} \quad (4.35)$$

$$\text{where } s' = \left(\frac{ds}{d\xi}\right)$$

$$\int d\xi = \int \frac{ds}{\sqrt{B_1 + B_2s + B_3s^2 + B_4s^3}} \quad (4.36)$$

$$\xi \equiv x - ct = \int \frac{ds}{\sqrt{B_1 + B_2s + B_3s^2 + B_4s^3}} \quad (4.37)$$

$$x - ct = \int \frac{ds}{\sqrt{(s-P)(s-Q)(s-T)}} \quad (4.38)$$

where P, Q, T are the distinct roots of the cubic equation,

$$B_1 + B_2s + B_3s^2 + B_4s^3 = 0.$$

where

$$P + Q + T = -\frac{B_3}{B_4} \quad (4.39)$$

$$PQ + PR + QT = -\frac{B_2}{B_4} \quad (4.40)$$

$$PQT = -\frac{B_1}{B_4} \quad (4.41)$$

Now substitute

$$s = T + (Q - P) \sin^2 \theta \quad (4.42)$$

$$x - ct = \frac{2}{\sqrt{P - T}} \int_0^\theta \frac{d\theta}{\sqrt{1 - m \sin^2 \theta}} \quad (4.43)$$

where

$$m = \frac{Q - T}{P - T} \quad (4.44)$$

and

$$cn[(x - ct) \sqrt{\frac{P - T}{2}} | m] = \cos \theta \quad (4.45)$$

$$r(x - ct) = Q - (Q - T) cn^2[(x - ct) \sqrt{\frac{P - T}{2}} | m] \quad (4.46)$$

and the phase can be computed for (4.19)

4.2 Limiting Behaviors of the Traveling wave.

In this section, it will be shown that the traveling wave is an intermediate wave between solitary wave and sinusoidal wave, depending on the value of the parameter m . On

Setting

$$Q - T = 2a \quad (4.47)$$

so that

$$m = \frac{2a}{P - T} \quad (4.48)$$

so that

$$\lim_{a \rightarrow 0} m = 0 \quad (4.49)$$

Also it is known from the theory of elliptic functions that

$$\lim_{m \rightarrow 0} \operatorname{cn}(\varphi | m) = \frac{1}{\cosh \varphi} \quad (4.50)$$

Thus in the limiting situation as $m \rightarrow 0$ one gets the sinusoidal wave

$$r(x - ct) \sim Q - 2a \cos^2[(x - ct) \sqrt{\frac{P - T}{2}}] \quad (4.51)$$

as $a \rightarrow 0$

On the other hand the solitary wave limit requires the two simple zeros at $s = Q, T$ to coincide to form a zero of multiplicity two. To accomplish this, one can let $P \rightarrow Q -$ (at fixed T) and this implies that $m \rightarrow 1 -$. Once again from the theory of elliptic functions,

$$\lim_{m \rightarrow 1-} \operatorname{cn}(\varphi | m) = \frac{1}{\cosh \varphi} \quad (4.52)$$

which leads to

$$r(x - ct) \rightarrow Q - \frac{(Q - T)}{\cosh^2[(x - ct) \sqrt{\frac{P - T}{2}}]} \quad (4.53)$$

as $Q \rightarrow P$, which gives the required solitary wave in the other limiting situation.

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